

AI Readiness Hackathon – KSA Final Submission Template

Team name: The Auditeers

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Solution name: ClinAI-Audit :AI Readiness & Critical Reasoning Simulator

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1. Introduction

Artificial Intelligence (AI) is increasingly utilized across healthcare to process complex clinical data and generate treatment recommendations. However, in clinical nursing and medical practice, the uncritical adoption of AI outputs introduces a critical vulnerability: Automation Bias—the tendency of healthcare trainees to blindly accept algorithmic suggestions without independent clinical validation. Addressing this bias during academic and residency stages is vital: timely intervention trains practitioners to critically audit AI, preserving clinical autonomy and protecting patient safety. ClinAI-Audit directly bridges technical AI execution with national regulatory frameworks. By mapping our interactive auditing platform to the AI Readiness Framework dimensions (data governance, model validation, clinical safety guardrails, and human-in-the-loop ethics), we ground our solution in national strategic requirements rather than treating it as a standalone gadget. This directly advances Saudi Arabia’s Vision 2030 goals for a safe, digitally mature, and ethically governed healthcare sector.

2- Description of the Use Case, Policy Gaps, and Solution

The Problem & Current Policy Gap: Current healthcare simulation tools focus exclusively on traditional manual diagnosis or passively display AI outputs without testing student vigilance. **Current Policy Gap:** While national and international frameworks (such as SDAIA AI Ethics Principles and WHO Guidance on AI in Health) advocate for a Human-in-the-Loop approach in clinical decisions, there is a total lack of regulatory or educational metrics to test and measure a practitioner's capability to audit, detect, and override AI hallucinations before clinical execution.

Our Solution: ClinAI-Audit is an interactive simulation platform backed by a Retrieval-Augmented Generation (RAG) pipeline connected to an authentic Saudi Knowledge Base

(SDAIA, MOH, PDPL). The platform presents virtual patient scenarios containing deliberate algorithmic errors and contraindications (e.g., prescribing a drug that conflicts with an allergy documented in the patient's record). Instead of accepting the recommendation, the student acts as a Critical Auditor, rejecting faulty steps and providing a documented clinical rationale. The system enforces MOH safety protocols (e.g., The 5 Rights of Medication Administration, Double-Check Protocol) and calculates a real-time AI Readiness Score. This bridges the governance gap by providing an objective metric evaluating digital vigilance and clinical reasoning prior to real-world deployment.

3. Mapped documents

Y.3172 Node PDF	Platform Application & Technical Role PDF	Mapped Knowledge Base (KB) Documents & Saudi Regulations DOCX+ 1
SRC (Data Source)	Virtual patient medical records (medical history, vital signs, lab results, documented allergies).	MOH Guidelines & SCFHS SOPs: Clinical Practice Guidelines & Patient Safety Protocols.
C (Collector)	Consolidates all virtual patient record parameters into a unified simulation view.	CST & MOH Regulations: Cloud Data Residency, Availability, and Reliability Guidelines.
PP (Pre-Processing)	Anonymizes virtual patient identifiers and encrypts sensitive records prior to processing.	Saudi PDPL (Royal Decree No. M/19):Article 4 (Data Minimization), Article 7 (Anonymization & De-identification), Article 12 (Confidentiality).
M (Model)	LLM engine generating treatment plans pre-programmed with deliberate AI Hallucinations & Traps.	SDAIA AI Ethics: Principle 3 (Fairness & Bias Mitigation), Principle 4 (Transparency & Explainability).
P (Policy Node)	Human-in-the-Loop Audit: Routes AI recommendations to the student to evaluate and submit clinical rationale. Halts	SDAIA AI Ethics: Principle 1 (Human Oversight), Principle 2 (Accountability). MOH Rules: Rule 1 (5 Rights), Rule 2

	plan if student falls into Automation Bias.	(Double-Check), Rule 3 (Clinical Reasoning Over Automated Output).
D(Distribution)	Evaluates student performance and computes the AI Readiness Score based on clinical accuracy and vigilance.	AI Readiness Framework 2.0: Performance analytics, risk scoring, and digital vigilance benchmarking.
SINK (Data Sink)	Displays detailed performance feedback on the Student Dashboard and Faculty Analytics Dashboard.	SCFHS & University Standards: Academic evaluation frameworks, competency tracking, and clinical licensing readiness.

4- Evaluation Scenarios & Controversy Resolution

Step 1: Deployment & Institutional Adoption

ClinAI-Audit is successfully deployed across medical and nursing simulation centers (such as King Saud bin Abdulaziz University for Health Sciences). The platform is actively used by nursing students to evaluate AI-generated clinical plans, successfully measuring and raising their baseline AI Readiness Scores from 85% to 92% across high-risk patient scenarios.

Step 2: Commercial Intrusion & Data Privacy Controversy (Controversy Scenario)

After 6 months of national adoption, monitoring agencies discover that the application service provider introduced targeted commercial promotions and third-party monetization within the mobile interface. Student performance metrics—such as audit failure rates, diagnostic hesitation times, and skill deficits—were being processed to push targeted pharmaceutical advertising and paid private certification courses directly to students.

Step 3: Policy-Driven Remediation via Knowledge Base Guardrails

This incident escalates the case from a basic terms-of-service dispute into a critical AI Ethics, Data Privacy, and Healthcare Governance Breach. Citing the pre-agreed regulatory rules embedded within our Knowledge Base (KB), ClinAI-Audit automatically triggers the following corrective enforcement actions:

1. Mandatory PDPL Data Isolation: Pursuant to Saudi Personal Data Protection Law (PDPL) Articles 4 and 12, all student clinical performance and behavioral audit data are strictly classified as sensitive educational metrics. The system immediately enforces purpose-limitation rules, blocking all commercial profiling and halting in-app targeted promotions.
2. Automated System Freeze & Revocation: Under SDAIA AI Ethics Principles (Principle 1: Human Oversight & Principle 2: Accountability), the Policy Node (Node P) automatically freezes commercial API data exchanges and revokes unauthorized third-party distribution channels (Node D).
3. National Regulatory Escalation: The compliance breach is automatically logged and reported to national oversight authorities (SDAIA and MOH) for official audit and remediation under pre-agreed national rules rather than ad hoc resolution.

1: Video:

https://drive.google.com/file/d/1nU1uS_c-W7F4I_bWJj61lggGQU7wBCtW/view?usp=sharing

2: GitHub code:

3: Knowledge base:

https://ksauhsedu-my.sharepoint.com/:w:/g/personal/alruwayshid30550_ksauhs_edu_sa/IQB7NJSUqt1vQKiW4tPeQQBKAQZDoX4YOoNRfQncT5xC798?e=w3Keia

